Lecture 21

6th Nov

ω : n^{th} primitive root of unity

$$\omega^n = 1$$

$$\omega^i \neq 1 \quad \text{for} \quad 1 \leq i < n$$

Fast Fourier Transform (FFT)

$$\begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega & \omega^2 & \cdots & \omega^{n-1} \\ 1 & \omega^2 & \omega^4 & \cdots & \omega^{2(n-1)} \\ \vdots & & & & \\ 1 & \omega^{n-1} & \omega^{(n-1)2} & \cdots & \omega^{(n-1)^2} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{n-1} \end{bmatrix} = \begin{bmatrix} P(1) \\ P(\omega) \\ \vdots \\ P(\omega^{n-1}) \end{bmatrix}$$

$$\left[V(\omega)\right]^{-1} = \frac{1}{n} V\left(\frac{1}{\omega}\right)$$

$FFT(n, a_0, a_1, \ldots, a_{n-1}, \omega, \text{var } V)$

begin

if $n = 1$ then

$V[0] = a_0$

else

$FFT(n/2, a_0, a_2, \ldots, a_{n-2}, \omega^2, U)$

$FFT(n/2, a_1, a_3, \ldots, a_{n-1}, \omega^2, W)$

for $j = 0$ to $n/2 - 1$ do

$V[j] = U[j] + \omega^j W[j]$

$V[j + n/2] = U[j] - \omega^j W[j]$

end

$(0,1,2,3,4,5,6,7)$

$P_{0,1,2,3,4,5,6,7}(1,\omega,\omega^2,\dots,\omega^7)$

$P_{0,2,4,6}(1,\omega^2,\omega^4,\omega^6)$ $\qquad$ $P_{1,3,5,7}(\omega,\omega^3,\omega^5,\omega^7)$

$P_{0,4}(1,\omega^4)$ $\qquad$ $P_{2,6}(1,\omega^4)$

$P_0(1) = 0$ $\qquad$ $P_4(1) = 4$

eg: $P_{2,6}(\omega^2,\omega^6) = (8,-4)$

$2\cdot(\omega^2) + 6\cdot(\omega^6)$